Fixed-base N53 S3 root-batch result

Nine verified single-target pairs, 2026-10-12

Question. Does batching inversions in the second S3 root call reduce charged rank and target work under the exact fixed K220 base and rank stream? Curve EC1N53Ce0hb097de99be9a, subgroup order 21,044,858,204,113. The frozen source, public inputs and binary hashes are in `FROZEN.json`; raw runs in `runs/`; native exact/replay verdict in `ANALYSIS.json` and aggregate in `RESULT_DATA.json`.

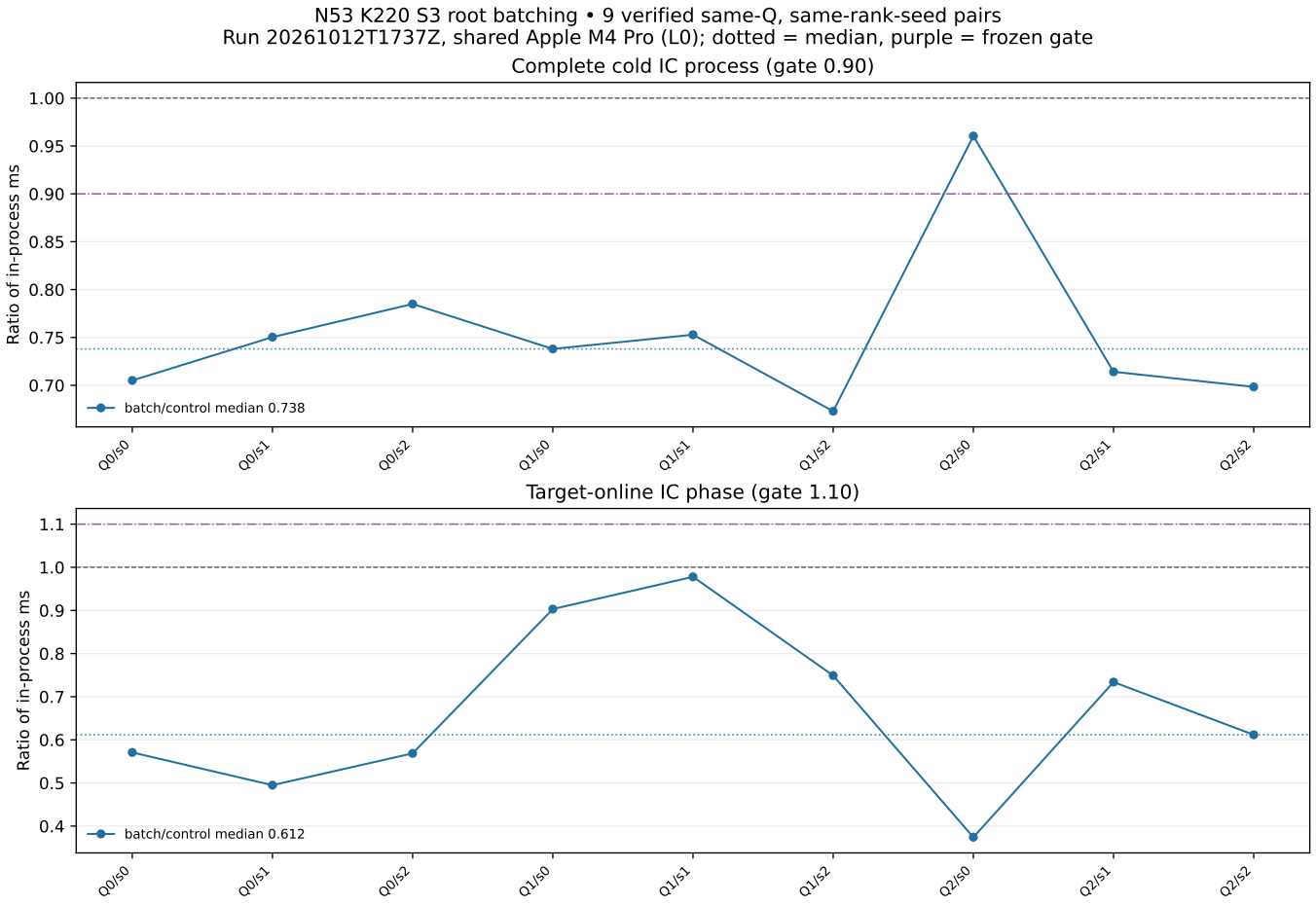
Verified result. All 27 valid fresh processes completed; all nine IC pairs reached rank 220/220 with 220 independent relations each. Native replay checked each relation, factor log and target scalar. The control and batch arms retain the same base, selected rank trace, root table, consumed roots and target witness. Exact and inversion gates passed.

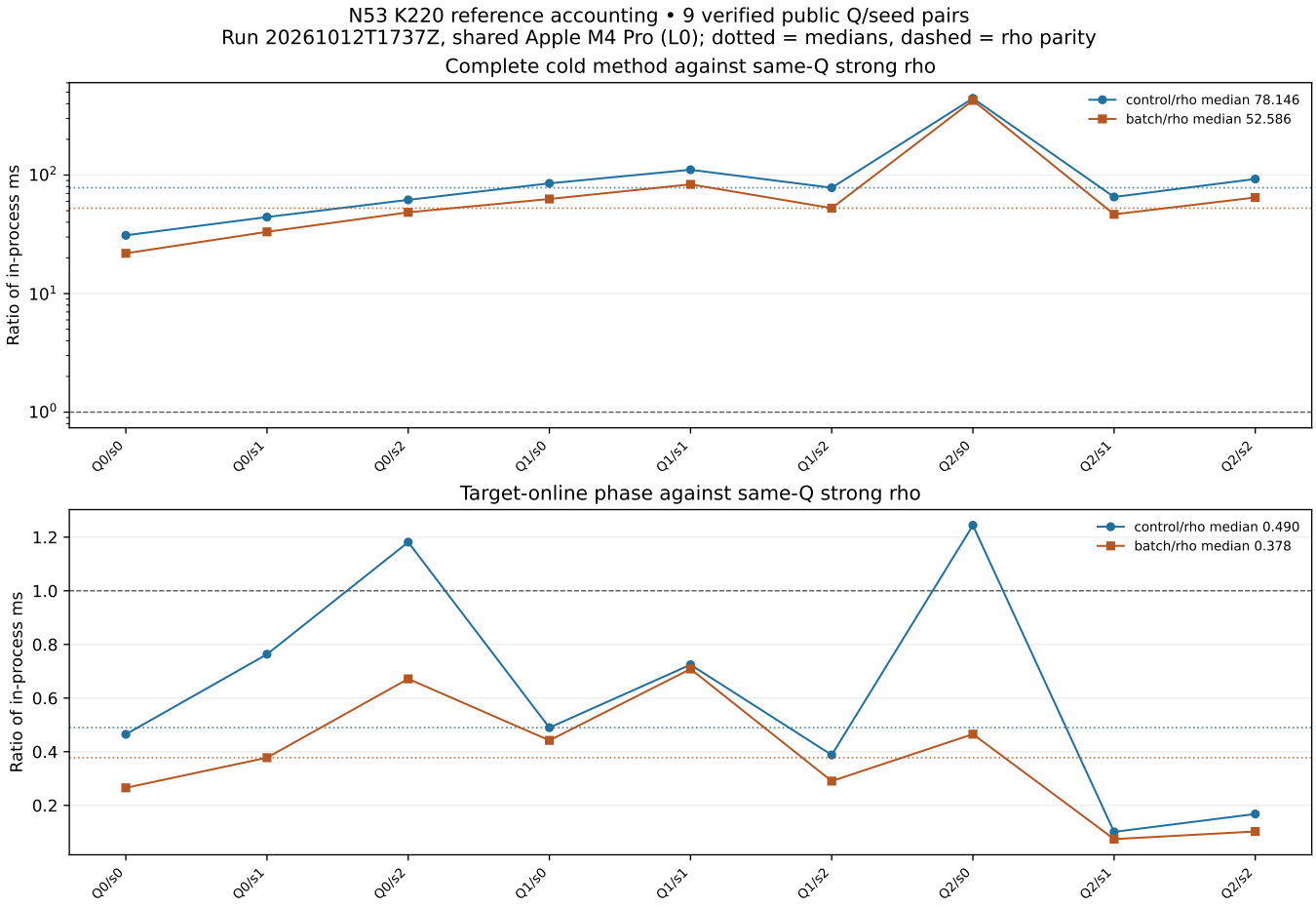
Same-Q ratio, in-process ms	Median	95% block interval	Scope
Control/strong rho, cold	78.146	[44.220, 92.602]	L0 diagnostic
Batch/strong rho, cold	52.586	[33.179, 64.668]	L0 diagnostic
Batch/control, cold	0.738	[0.714, 0.750]	L0 diagnostic
Control/strong rho, online	0.490	[0.168, 0.763]	L0 diagnostic
Batch/strong rho, online	0.378	[0.103, 0.442]	L0 diagnostic
Batch/control, online	0.612	[0.569, 0.903]	L0 diagnostic

The candidate/control descriptive gates were cold ≤ 0.90 and online ≤ 1.10, both met. Bootstrap intervals come from 10,000 deterministic resamples of three public-target blocks; each block retains its three rank seeds. The host was a shared Apple M4 Pro, macOS 26.6, with no core isolation. Rho uses the same Q, strong rung 3, 32 lanes and DP4.

Charged root primitives, nine IC cells. Control versus batch: 80,790,012 versus 763,167 inversions (ratio 0.009446); 1,211,850,180 versus 902,063,394 field multiplication calls (0.744369); 242,370,036 versus 39,684,684 squaring calls (0.163736). Extra prefetched roots are charged. This is engineering with unchanged relations, not a measured generic exponent change. The complete-operation unit $S = \mathrm{GAE}/\sqrt{r}$ and ratio to its $\sqrt{\pi/(2 \cdot 106)}$ floor remain uncalibrated. The cold IC/rho ratios are above 1 in every pair. Host-isolated timing and second-host replay remain open.

Custody and validation. The executable source commit was `00a8da2d`; public inputs and binary hashes were pushed in `576f7087b` before timing. A sandboxed `time -l` measurement-wrapper attempt returned incomplete statuses and is preserved separately; the valid run used the same source, inputs and schedule outside that sandbox. Final validation: 4,000 release library tests, 22 release example tests and four targeted scoreboard/site tests passed. The full site suite has one unrelated baseline error: the preverified cover report names a registry digest that differs from the unchanged registry on `origin/main`. The canonical scoreboard and timeline carry only L0 diagnostic points. A ledger promotion remains pending common-S calibration and an admitted isolated reference.


N53 K220 S3 root batching • 9 verified same-Q, same-rank-seed pairs
Run 20261012T1737Z, shared Apple M4 Pro (L0); dotted = median, purple = frozen gate
Complete cold IC process (gate 0.90)
Ratio of in-process ms
1.00
0.95
0.90
0.85
0.80
0.75
0.70
batch/control median 0.738
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2
Target-online IC phase (gate 1.10)
Ratio of in-process ms
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
batch/control median 0.612
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2


N53 K220 reference accounting • 9 verified public Q/seed pairs
Run 20261012T1737Z, shared Apple M4 Pro (L0); dotted = medians, dashed = rho parity
Complete cold method against same-Q strong rho
control/rho median 78.146
batch/rho median 52.586
Ratio of in-process ms
10^2
10^1
10^0
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2
Target-online phase against same-Q strong rho
control/rho median 0.490
batch/rho median 0.378
Ratio of in-process ms
1.2
1.0
0.8
0.6
0.4
0.2
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2